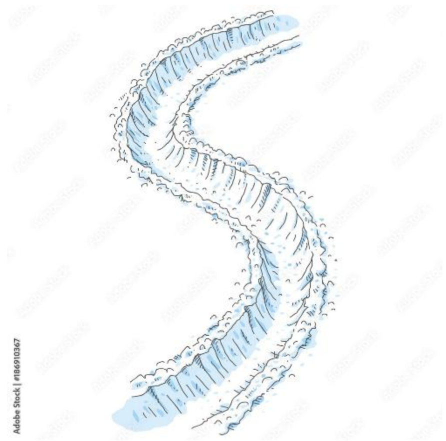
MTH 202 Chapter 5: Topics In Vector Calculus

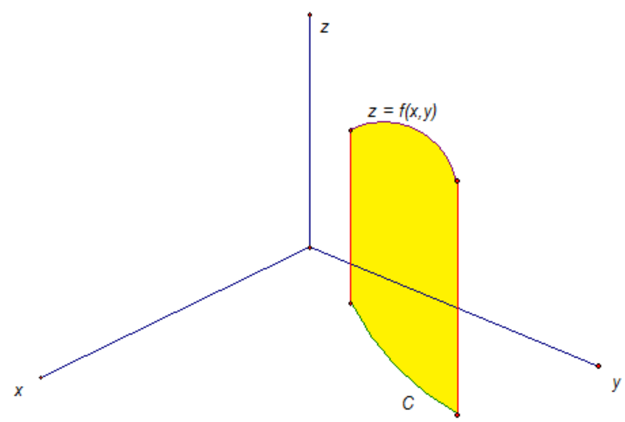
5.2.1 Line Integrals

Motivation: Suppose we just experienced a snowstorm and there is snow everywhere outside. In some spots the snow is higher and in some spots it is lower. You want to walk out into the snow, shoveling as you go, so that you can reach your car. In this scenario, how much snow do we actually have to shovel? This entirely depends on the path that we take to reach our car. How long the path is has an effect, but also the amount of snow that our specific path will take us through.



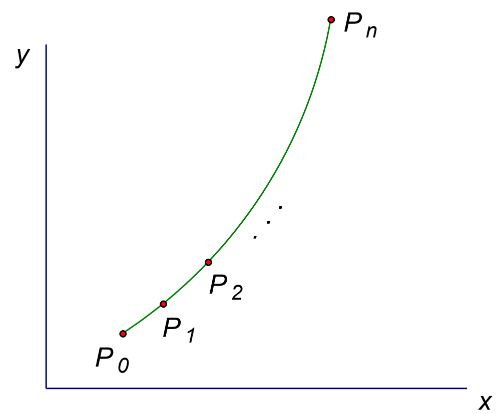
A <u>line integral</u> (or what is sometimes referred to as a <u>path integral</u>) captures this idea of accumulation of a specific quantity over a path. More specifically, suppose that we plan to move along a curve C in 3-space and $f(x,y)$ measures the amount of some quantity that occurs at each point along the path C. A line integral would help us to determine the accumulation of $f(x,y)$ over the curve C.

Exploration: Suppose that the continuous function $f(x,y)$ measures the amount of "quantity" that occurs at each point along a smooth plane curve C and we want to add up the amount of "quantity" that occurs as we move along the curve. *Note: This is the same as saying that we want the area of the "curtain" formed by* $f(x,y)$ *over C.* A possible picture of the situation is shown below (the "curtain" is in yellow).



In the picture the green curve C lies completely on the xy-plane and the purple curve $z = f(x,y)$ is the result when only points on C are plugged into $f(x,y)$.

To estimate the area of the "curtain" we partition $[a,b]$ into n subintervals of the form $[t_{i-1}, t_i]$ each of width Δt.
We can use this partition to partition the curve C, if we plot each point $P_i : (x_i, y_i)$ on C



Then, we let $\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$ be the length of the line segment from P_{i-1} to P_i.
We select a sample point (x_i^*, y_i^*) from each interval, and we then $f(x_i^*, y_i^*)$ acts as a representative height of the sheet at this point.

Then, an approximation of our "curtain" area is: $A \approx \sum_{i=1}^{n} f(x_i^*, y_i^*) \Delta s_i$

This leads to the following definition.

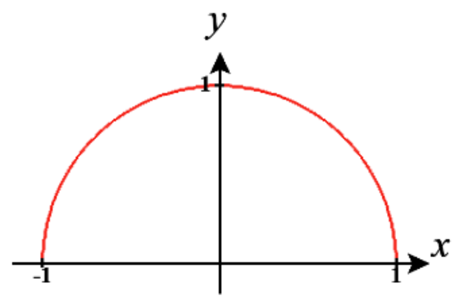
Definition: If f is defined on a smooth curve C, then the <u>line integral of f along C</u> is

$$\int_C f(x,y)\,ds = \lim_{n\to\infty}\sum_{i=1}^{n} f\left(x_i^*, y_i^*\right)\Delta s_i$$

provided this limit exists.

Note: In later sections we will want to extend this idea to adding up quantities as we move across a surface.

<u>Example 1:</u> Suppose a wire can be modelled by the upper half of the unit circle given by $x^2+y^2=1$. If the linear density of this wire is measured by $f(x,y)=2+x^2y$ for any point on the wire, then set up an integral which represents the mass of this wire.



We know that the mass of the wire is given by $M=\int_C\left(2+x^2y\right)ds$

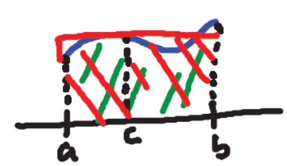
where C is the upper half of the unit circle.

While we may follow the logic of how this integral was constructed it is strange to think about how one would evaluate an integral of this type.

Exploration: We know from chapter 2 that we can parameterize C by $\mathbf{r}(t)=\langle x(t),y(t)\rangle$ where $a\le t\le b$.
Suppose we do this and observe the ith segment of the curve C. We see that from section 2.3 the arc length of this segment is given by:

$$\Delta s_i=\int_{t_{i-1}}^{t_i}\left\|\mathbf{r}'(t)\right\|dt$$

However, we may recall the Mean Value Theorem for Integrals (from Calculus I) which states that for a continuous function



$$\frac{1}{b-a}\int_a^b f(t)\,dt=f(c)\quad\Rightarrow\quad\int_a^b f(t)\,dt=f(c)(b-a)\ \text{for some } c \text{ in } [a,b].$$

Therefore, by the Mean Value Theorem for Integrals we know that there exists a t_i^* in $\left[t_{i-1},t_i\right]$ such that

$$\Delta s_i=\int_{t_{i-1}}^{t_i}\left\|\mathbf{r}'(t)\right\|dt=\left\|\mathbf{r}'\left(t_i^*\right)\right\|\left(t_i-t_{i-1}\right)=\left\|\mathbf{r}'\left(t_i^*\right)\right\|\Delta t$$

So, $\int_C f(x,y)\,ds=\lim_{n\to\infty}\sum_{i=1}^{n} f\left(x_i^*,y_i^*\right)\Delta s_i=\lim_{n\to\infty}\sum_{i=1}^{n} f\left(x_i^*,y_i^*\right)\left\|\mathbf{r}'\left(t_i^*\right)\right\|\Delta t=\int_a^b f\left(x(t),y(t)\right)\left\|\mathbf{r}'(t)\right\|dt$.

We have transformed the original line integral into an integral which is completely in terms of *t*!

Fact: If $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ where $a \le t \le b$ represents the parameterization of *C* then,

$$\int_C f(x,y)\,ds = \int_a^b f(x(t), y(t)) \|\mathbf{r}'(t)\|\,dt$$

One important item to note is that the value of the line integral does NOT depend on the parameterization of *C* (the parameterization was $\mathbf{r}(t)$ above) provided *C* is traversed exactly once as *t* increases from *a* to *b*.

Remark: We can further simplify the above formula by noticing that $\|\mathbf{r}'(t)\| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ and therefore we have:

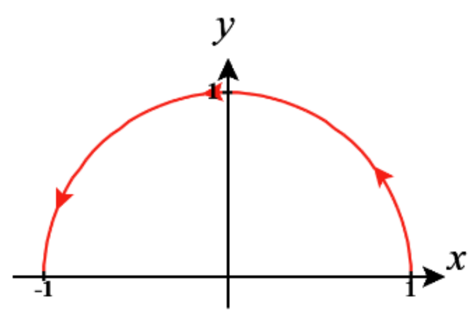
$$\int_C f(x,y)\,ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt.$$

<u>Example 1 (cont.):</u> Suppose a wire can be modelled by the upper half of the unit circle given by $x^2 + y^2 = 1$. If the linear density of this wire is measured by $f(x,y) = 2 + x^2 y$ for any point on the wire, then set up and evaluate an integral which represents the mass of this wire.

We already established from the first part of this example that $M = \int_C (2 + x^2 y)\,ds$

Now we need to parameterize *C* so that we can get this integral into a state where we can compute its value.

One possible parameterization for the top half of the unit circle is $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$ where $0 \le t \le \pi$ (*Note: Another possible parameterization is* $\mathbf{r}(t) = \langle t, \sqrt{1-t^2} \rangle$ *where* $-1 \le t \le 1$, *but this parameterization makes evaluation messier*).



So, we have that

$$M = \int_C (2 + x^2 y)\,ds = \int_0^\pi (2 + \cos^2(t)\sin(t)) \sqrt{(-\sin(t))^2 + (\cos(t))^2}\,dt = \int_0^\pi (2 + \cos^2(t)\sin(t))\,dt$$

$$= 2t - \frac{1}{3}\cos^3(t)\Big|_0^\pi = \left(2\pi + \frac{1}{3}\right) - \left(\frac{-1}{3}\right) = 2\pi + \frac{2}{3}.$$

Note: As we see in the above examples, one of the biggest challenges in evaluating a line integral can be finding a suitable parameterization for C.

Extensions to 3-Dimensions
All of the notions above extend easily to three dimensions. In particular if *C* is a smooth space curve given by $\mathbf{r}(t)=\langle x(t),y(t),z(t)\rangle$ where $a\le t\le b$ and f is a three variable function that is continuous on some region containing *C*, then the line integral of f along *C* is

$$\int_C f(x,y,z)\,ds=\int_a^b f(x(t),y(t),z(t))\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt$$

Example 2: Suppose you have a pipe in 3-space that follows the path of the curve *C* parameterized by $\mathbf{r}(t)=\langle\cos(t),\sin(t),t\rangle$ where $0\le t\le 2\pi$. All suppose that the pipe varies in size across length and the cross-sectional area at any point can be given the function $f(x(t),y(t),z(t))=\pi(t+1)^2$. Set up and evaluate an integral which represents the volume of water that can exist within this pipe.

To get the volume we can see that we just want to add up $\left(\text{cross-sectional area}\times ds\right)$.

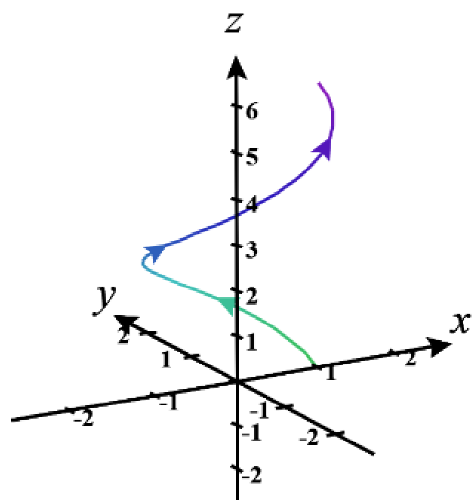
Therefore, the volume is given by

$$V=\int_C f(x,y,z)\,ds=\int_C f(x(t),y(t),z(t))\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt$$

$$=\int_0^{2\pi}\pi(t+1)^2\sqrt{\sin^2(t)+\cos^2(t)+1}\,dt=\int_0^{2\pi}\sqrt{2}\pi(t+1)^2\,dt$$

$$=\frac{\sqrt{2}\pi}{3}(t+1)^3\Big|_0^{2\pi}=\left(\frac{\sqrt{2}\pi}{3}(2\pi+1)^3-\frac{\sqrt{2}\pi}{3}\right)$$

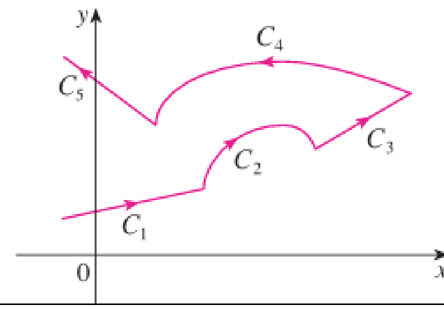
$$=\frac{\sqrt{2}\pi}{3}\left[(2\pi+1)^3-1\right]$$



Note: In the special case that $f(x,y,z)=1$, *we note that* $\int_C ds=\int_a^b\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt$ *which means that in this case the line integral of* f *along C is equal to the arc length of C!*

Definition: We say that the curve C is a piecewise-smooth curve if C is the union of a finite number of smooth curves $C_1, C_2, \ldots, C_n$ where the initial point of C_{i+1} is the terminal point of C_i.

The picture on the right shows a function that is piecewise-smooth and is made up of 5 different curves.

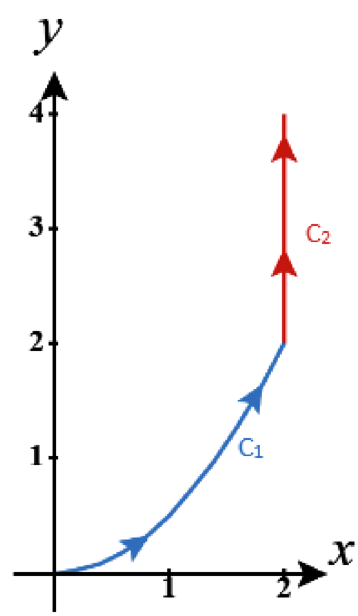


Fact: If the curve C is a piecewise-smooth curve, then we define the integral of f along C as the sum of the integrals of f along each of the smooth pieces of C.

If C is the union of a finite number of smooth curves $C_1, C_2, \ldots, C_n$ where the initial point of C_{i+1} is the terminal point of C_i, we write: $\int_C f(x,y)\,ds = \int_{C_1} f(x,y)\,ds + \int_{C_2} f(x,y)\,ds + \cdots + \int_{C_n} f(x,y)\,ds$

Example 3: Suppose that a curve C consists of the arc C_1 of the parabola $y = \frac{x^2}{2}$ from $(0,0)$ to $(2,2)$ followed by the line segment C_2 from $(2,2)$ to $(2,4)$. Evaluate $\int_C x\,ds$.

We first note that a parameterization for C_1 is $\mathbf{r}_1(t) = \left\langle t, \frac{t^2}{2} \right\rangle$ where $0 \le t \le 2$, and a parameterization for C_2 is $\mathbf{r}_2(t) = \langle 2, t \rangle$ where $2 \le t \le 4$.



So, we have that

$$\int_C x\,ds = \int_{C_1} x\,ds + \int_{C_2} x\,ds = \int_0^2 t\sqrt{1+t^2}\,dt + \int_2^4 2\sqrt{0+1^2}\,dt$$

$$= \frac{1}{3}\left(1+t^2\right)^{\frac{3}{2}}\Big|_0^2 + 2t\Big|_2^4 = \frac{5\sqrt{5}-1}{3} + 4$$

$$= \frac{5\sqrt{5}+11}{3}$$

Now, we will go over two different types of line integrals which are important in applications.

Definition:

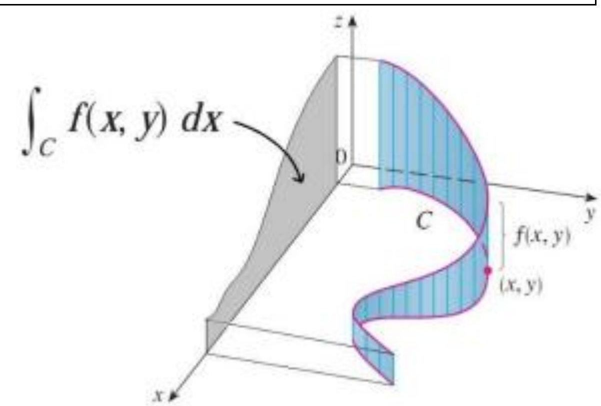
1. If we replace Δs_i with Δx_i in the first definition of this lecture, we obtain the <u>line integral of f along C with respect to x</u>: $\int_C f(x,y)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*, y_i^*)\Delta x_i$.
2. If we replace Δs_i with Δy_i in the first definition of this lecture, we obtain the <u>line integral of f along C with respect to y</u>: $\int_C f(x,y)\,dy = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*, y_i^*)\Delta y_i$.

Note: To avoid confusion, we call the line integral defined in our first definition of this lecture the ***<u>line integral with respect to arc length</u>****.*

Fact: If C is a smooth curve with parameterization $\mathbf{r}(t) = \langle x(t), y(t)\rangle$ where $a \le t \le b$ and f is continuous on a region containing C, then

$$\int_C f(x,y)\,dx = \int_a^b f(x(t),y(t))\,x'(t)\,dt \quad \text{and} \quad \int_C f(x,y)\,dy = \int_a^b f(x(t),y(t))\,y'(t)\,dt$$

Note: If the line integral with respect to the arc length is the size of the "curtain" beneath $f(x,y)$, you can think of the line integral of f with respect to x or y as the size of the shadow (projection) in the x or y directions. The real significance of line integrals with respect to x, y, and z will be explored in the next section when we look at the relationship between line integrals and the notation of work.



Note: The above definition and fact extend easily to three dimensions where we also have a line integral of f along C with respect to z.

Note: In all cases of line integrals that we have seen (with respect to x, y, z, or arc length) we solve these integrals by parametrizing C in terms of t, and then calculating the integral (and its components) in terms of t.

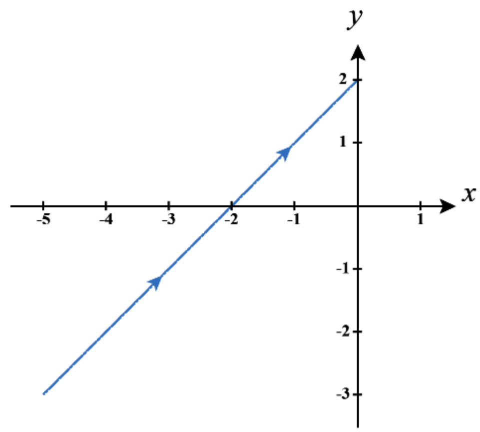
Notation: Often times line integrals with respect to x and y occur together. When this happens, it is customary to abbreviate as follows: $\int_C P(x,y)\,dx + \int_C Q(x,y)\,dy = \int_C P(x,y)\,dx + Q(x,y)\,dy$. An analogous abbreviation is used for three line integrals with respect to x, y, and z.

Example 4: Evaluate $\int_C y^2 dx + xdy$ where C is the line segment from $(-5,-3)$ to $(0,2)$.

We could find a parameterization for C by using techniques of basic algebra, but it can be faster to find a parameterization through the techniques of chapter 2.

In particular in chapter 2 we learned that a vector equation for the line connecting the terminal points of the position vectors $\mathbf{r}_0 = \langle -5,-3 \rangle$ and $\mathbf{r}_1 = \langle 0,2 \rangle$ is

$\mathbf{r}(t) = \mathbf{r}_0 + t(\mathbf{r}_1 - \mathbf{r}_0) \Leftrightarrow \mathbf{r}(t) = \langle -5,-3 \rangle + t\langle 5,5 \rangle \Leftrightarrow \mathbf{r}(t) = \langle -5+5t, -3+5t \rangle$ where $0 \le t \le 1$.



So, using $\int_C f(x,y)\,dx = \int_a^b f(x(t),y(t))\,x'(t)\,dt$ and

$\int_C f(x,y)\,dy = \int_a^b f(x(t),y(t))\,y'(t)\,dt$, we have that:

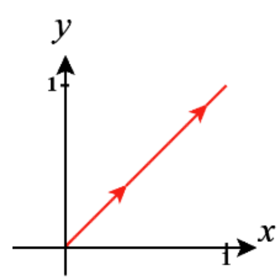
$$\begin{aligned}\int_C y^2 dx + xdy &= \int_C y^2 dx + \int_C x\,dy \\ &= \int_0^1 (-3+5t)^2 (5)\,dt + \int_0^1 (-5+5t)(5)\,dt \\ &= \int_0^1 \left(125t^2 - 125t + 20\right) dt = \frac{-5}{6}.\end{aligned}$$

Question: Would our answer in the above example have changed if we found a parameterization from (0,2) to (-5,-3)? The next example addresses this question.

Notation: Suppose that C is a curve with initial point A and terminal point B. The curve $-C$ denotes the curve consisting of the same points as C, but with opposite orientation (from initial point B to terminal point A).

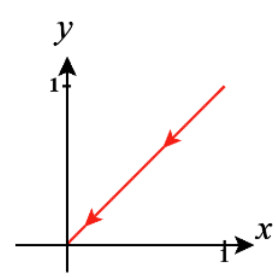
Example 5:

a) Suppose that C is given by $\mathbf{r}(t) = \langle t,t \rangle$ where $0 \le t \le 1$. Find $\int_C xy\,dy$.



We calculate that $\int_C xy\,dy = \int_0^1 t^2 dt = \frac{1}{3}t^3\Big|_0^1 = \frac{1}{3}$.

b) Suppose that C' is given by $\mathbf{r}(t) = \langle 1-t, 1-t \rangle$ where $0 \le t \le 1$. Find $\int_{C'} xy\,dy$.



We calculate that $\int_C xy\,dy = -\int_0^1 (1-t)^2\,dt = -\int_0^1 \left(1-2t+t^2\right) dt = \left[-\left(t - t^2 + \frac{1}{3}t^3\right)\right]_0^1 = \frac{-1}{3}$.

c) Notice that $C' = -C$. What would you speculate is the relationship between $\int_C f(x,y)\,dy$ and $\int_{-C} f(x,y)\,dy$?

Based upon the answers to (a) and (b) we would speculate that $\int_{-C} f(x,y)\,dy = -\int_C f(x,y)\,dy$.

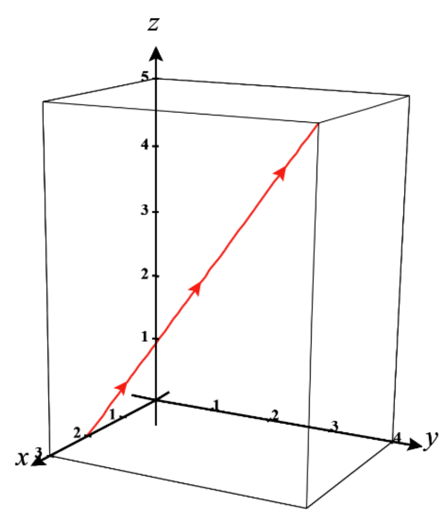
d) What is the relationship between $\int_C f(x,y)\,ds$ and $\int_{-C} f(x,y)\,ds$.

We know from this class that the line integral with respect to arc length does not depend on the parameterization of C provided C is traversed exactly once. So, we have that $\int_C f(x,y)\,ds = \int_{-C} f(x,y)\,ds$.

Fact: For line integrals that exist we have that $\int_{-C} f(x,y)\,dy = -\int_C f(x,y)\,dy$, $\int_{-C} f(x,y)\,dx = -\int_C f(x,y)\,dx$, and $\int_C f(x,y)\,ds = \int_{-C} f(x,y)\,ds$. These facts also extend to three dimensions.

Example 6: Suppose that C is the line segment from $(2,0,0)$ to $(3,4,5)$. Evaluate $\int_C y\,dx + z\,dy + x\,dz$.

We recall that there are actually three line integrals to evaluate here. In particular, $\int_C y\,dx + z\,dy + x\,dz = \int_C y\,dx + \int_C z\,dy + \int_C x\,dz$.



If we let $\mathbf{r}_0 = \langle 2,0,0\rangle$ and $\mathbf{r}_1 = \langle 3,4,5\rangle$, we know that a parameterization on the line segment from $(2,0,0)$ to $(3,4,5)$ is:

$\mathbf{r}(t) = \mathbf{r}_0 + t(\mathbf{r}_1 - \mathbf{r}_0) \Leftrightarrow \mathbf{r}(t) = \langle 2,0,0\rangle + t\langle 1,4,5\rangle \Leftrightarrow \mathbf{r}(t) = \langle 2+t, 4t, 5t\rangle$

where $0 \le t \le 1$.

So, we have that the line segment is given by the parametric equations:

$x(t) = 2+t \quad y(t) = 4t \quad z(t) = 5t$.

These parametric equations imply that $dx = dt \quad dy = 4dt \quad dz = 5dt$.

So, we have that

$$\int_C y\,dx + \int_C z\,dy + \int_C x\,dz = \int_0^1 4t\,dt + \int_0^1 5t\cdot 4\,dt + \int_0^1 (2+t)\cdot 5\,dt = \int_0^1 (4t + 20t + 10 + 5t)\,dt$$

$$= \int_0^1 (10 + 29t)\,dt = \left(10t + \frac{29t^2}{2}\right)\Bigg|_0^1 = 10 + \frac{29}{2} = \frac{49}{2}.$$